JOHANSSON ENGINEERING

Supervision Program — Gothenburg Workshop

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Revision	A
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Status	CURRENT
Author	N. Johansson
Project	JDS-PRJ-MEC-002
Client	Example Workshop AB
Site	Gothenburg Workshop
Program ID	SP-[NNN]
Source	JDS-LOG-MEC-006_example-inventory.md

1. Document Chain

```
INVENTORY  →  [PROGRAM]  →  ROUND  →  REVIEW
(Step 1)       (you are      (next)
                here)
```

Source document: JDS-LOG-MEC-006_example-inventory.md

2. Regulatory Basis

This program satisfies the requirements of **AFS 2017:3** (consolidated with AFS 2019:1, AFS 2020:10, AFS 2022:2):

REQUIREMENT	AFS 2017:3	STATUS
Documented supervision routines	4 Kap. §17	This document
Mandatory minimum checks (6 points)	2 Kap. §6	Section 5
Annual evaluation and revision	4 Kap. §17	Section 12
Assigned responsible person	4 Kap. §17	Section 9
Coordination person	4 Kap. §14	Section 9
Equipment register	4 Kap. §15	Section 3
Monitoring requirements	4 Kap. §16	Section 4
Recurring inspection schedule	5 Kap., Bilaga 1	Section 6
Deviation reports	4 Kap. §19	Section 10
Lifetime journal	4 Kap. §18	Section 11

Warning: Non-compliance sanction: **10,000–100,000 SEK** for operating Class A/B equipment without documented supervision routines (4 Kap. §17).

3. Equipment Register (4 Kap. §15)

Register of all pressurised equipment at Gothenburg Workshop:

VESSEL ID	DESCRIPTION	LOCATION	CLASS	MEDIUM	INSPECTOR
PV-001	Main air receiver	Compressor room	**A**	compressed air	Accredited body (Type A)
PV-002	Workshop air receiver	Workshop B	**Exempt (air/N2)**	compressed air	N/A
PV-003	Steam boiler	Boiler house	**A**	steam	Accredited body (Type A)
PV-004	Refrigeration receiver	Engine room	**A**	ammonia	Accredited body (Type A)
PV-005	Hydraulic accumulator	Workshop A	**B**	hydraulic oil	Accredited body (Type A or B)
PV-006	Expansion vessel	Heating system	**Below threshold**	water	N/A
PV-007	Sandblast receiver	Yard	**Exempt (air/N2)**	compressed air	N/A

Class A: 3 vessels (accredited inspection body required) **Class B:** 1 vessels **Exempt (air/N2/refrigerant):** 2 vessels (Class B exempt per 4 Kap. §10) **Below threshold:** 1 vessels

4. Monitoring Requirements (4 Kap. §16)

Class A and B vessels must be **continuously monitored** (§16). The operator must be able to immediately reach the vessel and determine if it is safe to remain pressurised.

If periodic monitoring is used instead (per documented risk assessment), the following must be documented:

1. How operators are alerted to safety-related alarms
2. The response time (inställelsetid) for safety-related alarms

VESSEL ID	MONITORING TYPE	ALARM SYSTEM	RESPONSE TIME
PV-001	Continuous / Periodic	[Describe]	[Minutes]
PV-003	Continuous / Periodic	[Describe]	[Minutes]
PV-004	Continuous / Periodic	[Describe]	[Minutes]
PV-005	Continuous / Periodic	[Describe]	[Minutes]

5. Mandatory Supervision Checks (2 Kap. §6)

AFS 2017:3 §6 defines **six mandatory minimum checks**. Every supervision round must verify all six points:

#	§6 REQUIREMENT	CHECK	METHOD
1	Equipment functions satisfactorily	Pressure/temperature within range, no abnormal noise/vibration	Gauge reading, listening
2	No leaks have occurred	Check flanges, valves, fittings, weld joints	Visual walk-around
3	No harmful external or internal impact	Surface condition, insulation, supports, corrosion	Close visual inspection
4	No other faults or deviations	Safety devices, drainage, general condition	Functional check
5	Equipment correctly marked	Nameplates, valve tags, emergency stops legible	Visual check
6	Prescribed inspections carried out	Inspection certificate current, not overdue	Register review

Important: These six points are the **legal minimum** per 2 Kap. §6 and must be traceable in every supervision record.

6. Recurring Inspection Schedule (Bilaga 1)

6.1 Driftprov (Operational Test)

Base intervals per Bilaga 1, §1.4.1. Actual interval determined by accredited inspection body at each inspection.

VESSEL ID	CLASS	MEDIUM	BASE INTERVAL	MAX
PV-001	A	compressed air	4 years	4 years
PV-003	A	steam	2 years	4 years
PV-004	A	ammonia	2 years	4 years
PV-005	B	hydraulic oil	2 years	4 years

Extended interval (§1.4.2): Possible if safety equipment functioned at the two previous tests. Max 4 years.

Shortened interval (§1.4.3): If safety equipment did NOT function, next interval is **halved**.

6.2 Internal/External Examination (Class A only)

Condition-based interval determined by inspection body per Bilaga 1, §2.2:

INTERVAL	CONDITIONS
4 years	Base: not fire-affected, >5yr life, low crack risk, mild environment
2 years	4-year not met, but stable condition
1 year	2-year not met, but safe for 1 year
6 months	1-year not met, but safe for 6 months
6 years	After clean 4-year, no fatigue/creep
8 years	After 2 clean 4/6-year inspections
10 years	After 2 progressively longer, clean
12 years	Cisterns only, after clean 6-year

VESSEL ID	CURRENT INTERVAL	NEXT DUE
PV-001	[Per inspection body]	[Expiry month]
PV-003	[Per inspection body]	[Expiry month]
PV-004	[Per inspection body]	[Expiry month]

7. Supervision Round Schedule

7.1 Daily / Per-Shift Checks

Applies to: PV-001, PV-003, PV-004, PV-005

#	CHECK (§6 REF)	METHOD
1	Pressure gauge within normal range	Visual reading
2	Temperature within design limits	Visual reading
3	No audible leaks or unusual noise	Listening
4	Control system: no standing alarms	Control panel check

Performed by: [Name / role] **Record:** Shift log

7.2 Weekly Checks

Applies to: PV-001, PV-003, PV-004, PV-005

#	CHECK (§6 REF)	METHOD
1	No visible leaks at flanges, valves, fittings	Walk-around visual
2	Safety valves not gagged or blocked	Visual check
3	Drain valves functional (operate drain)	Manual operation
4	Condensate drainage working	Visual / operate trap

Performed by: [Name / role] **Record:** Weekly check sheet

7.3 Monthly Checks (Formal Supervision Round)

Applies to: PV-001, PV-003, PV-004, PV-005, PV-002, PV-007

#	CHECK (§6 REF)	METHOD
1	External surface: no corrosion, dents, cracks	Close visual
2	Insulation intact, no moisture indicators	Visual check
3	Support and foundation integrity	Visual check
4	Safety valve seal intact	Visual check
5	Vessel access clear and safe	Visual check

Performed by: [Name / role] **Record:** Supervision round record (JDS-TMP-LOG-006)

7.4 Quarterly Checks

Applies to: PV-001, PV-003, PV-004, PV-005

#	CHECK (§6 REF)	METHOD
1	Nameplate legible and not obscured	Visual check
2	Pressure switch function test	Simulate / trip test
3	Paint / coating condition	Visual check
4	Review open findings from previous rounds	Register review

Performed by: [Name / role] **Record:** Supervision round record (JDS-TMP-LOG-006)

7.5 Annual Checks

Applies to: all vessels

#	CHECK	METHOD
1	Safety valve function test (lift test)	On-line or bench test
2	Equipment register accuracy verification	Compare to physical
3	Competence records current	Record review
4	Formal inspection schedule review	Inspection plan review

Performed by: Program manager / competent person **Record:** Annual review record (JDS-TMP-LOG-007)

8. Per-Vessel Check Assignment

VESSEL ID	CLASS	DAILY	WEEKLY	MONTHLY	QUARTERLY	ANNUAL
PV-001	**A**	Yes	Yes	Yes	Yes	Yes
PV-002	**Exempt (air/N2)**	—	—	Yes	—	Yes
PV-003	**A**	Yes	Yes	Yes	Yes	Yes
PV-004	**A**	Yes	Yes	Yes	Yes	Yes
PV-005	**B**	Yes	Yes	Yes	Yes	Yes
PV-006	**Below threshold**	—	—	Yes	—	Yes
PV-007	**Exempt (air/N2)**	—	—	Yes	—	Yes

9. Personnel and Responsibilities (4 Kap. §14, §17)

9.1 Assigned Roles

ROLE	AFS REF	NAME	RESPONSIBILITY
Coordination person	4 Kap. §14	[Name]	Plans and coordinates all work on Class A/B equipment
Supervision responsible	4 Kap. §17	[Name]	Ensures supervision is carried out and documented
Daily/weekly supervisor	2 Kap. §6	[Name]	Performs daily and weekly checks
Monthly supervisor	2 Kap. §6	[Name]	Performs monthly formal rounds
Program manager	4 Kap. §17	[Name]	Annual evaluation, revision, findings management

9.2 Competence Requirements

All personnel must have documented competence per JDS-PRO-009. Competence records must be maintained and refreshed.

10. Deviation Reports (4 Kap. §19)

When Class A/B equipment is found to be damaged or deteriorated, a **deviation report** must be created containing:

#	REQUIRED CONTENT	DESCRIPTION
1	Damage/deterioration	What was found
2	How discovered	Which observation or check
3	Date of discovery	When it was found
4	Action needed	What must be done
5	Cause	Root cause (if not obvious)
6	Date of action	When the repair/fix was completed
7	Reporter	Who made the report

Deviation reports are managed per **JDS-PRO-008** (Corrective Action Procedure) and filed in the client's `findings/` folder.

Severity Classification

SEVERITY	DEFINITION	ACTION	TIMELINE
Critical	Immediate safety risk	Out of service, escalate	Immediate
Major	Will deteriorate to critical	Plan repair, close monitoring	30 days
Minor	Noted, no immediate risk	Monitor, plan maintenance	90 days
Observation	Worth monitoring	Note, observe trend	Next round

11. Lifetime Journal (4 Kap. §18)

Class A/B equipment with limited lifetime must have a **journal showing remaining lifetime**. If parts have different lifetimes, each part must be tracked separately.

VESSEL ID	LIMITED LIFETIME	JOURNAL REF	NOTES
PV-001	Yes / No / Unknown	[Ref]	
PV-003	Yes / No / Unknown	[Ref]	
PV-004	Yes / No / Unknown	[Ref]	
PV-005	Yes / No / Unknown	[Ref]	

Warning: Equipment that has reached its documented lifetime **may not remain pressurised** unless an analysis demonstrating extended lifetime has been conducted and documented by an accredited body.

12. Program Review (4 Kap. §17)

This program must be **evaluated and revised at least once per year** (4 Kap. §17). Additional review triggers:

- Equipment added, removed, or modified
- Operating conditions changed significantly
- Regulatory requirements changed
- Significant finding or incident occurred
- Revision inspection (revisionskontroll) performed

Next review due: 2027-01-31

13. Approval

Prepared by	[Name, role]
Reviewed by	[Name, role]
Approved by	[Operator/employer, role]
Approval date	2026-04-10
Next review due	2027-01-31

14. Next Step

```
INVENTORY  →  [PROGRAM]  →  ROUND  →  REVIEW
                (done)      (next)
```

To generate a supervision round record:

```
python3 scripts/jds-classify.py --round --from [this-file.md] --output [round-YYYY-MM-DD.md]
```

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-04-10	N. Johansson	Initial program — 7 vessels, compliant with AFS 2017:3 (verified against official PDF). Auto-generated from JDS-LOG-MEC-006_example-inventory.md